



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2024/25 END OF SEMESTER I EXAMINATIONS

PHYS-1101: MECHANICS AND THERMAL PROPERTIES OF MATTER

[100 marks]

DATE: 28th October 2024

TIME: 3hr: 9:00 - 12:00

INSTRUCTIONS

1. Write your **Registration Number** on each page of your answer sheets.
2. Start each question on a **NEW** page.
3. This paper contains **7 printed pages** and **5 QUESTIONS**, please check.
4. Answer **ALL** questions.
5. Show your work clearly.
6. Marks allocated for each question are indicated against it.
7. Round off your final answer to **2 decimal places**.

CONSTANT(S)

Acceleration due to gravity	g	9.8 m/s^2
Gravitation constant	G	$6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$
Radius of the Earth	R_E	$6.37 \times 10^6 \text{ m}$
Mass of the Earth	M_E	$5.98 \times 10^{24} \text{ kg}$
Latent Heat of fusion of water	L_w	$6.0 \times 10^3 \text{ J.kg}^{-1}$
Gas constant	R	$8.315 \text{ J mol}^{-1} \text{ K}^{-1}$
Speed of light (in vacuum)	c	$2.998 \times 10^8 \text{ m s}^{-1}$
Speed of sound in air	v_a	343 m/s
Speed of sound in water	v_w	1480 m/s
Specific heat capacity of water	c_w	$4.184 \times 10^3 \text{ Jkg}^{-1} \text{ K}^{-1}$
Latent heat of fusion of ice	L_{ice}	334000 Jkg^{-1}
Specific heat capacity of steam	c_s	$2.01 \times 10^3 \text{ Jkg}^{-1} \text{ K}^{-1}$
Latent heat of vaporisation of water,	L_w	$2.260 \times 10^6 \text{ Jkg}^{-1} \text{ K}^{-1}$
Specific heat capacity of ice	c_{ice}	$2200 \text{ Jkg}^{-1} \text{ K}^{-1}$
Room temperature	T_R	25°C
Coefficient of linear expansion of Lead.		$2.9 \times 10^{-5} \text{ K}^{-1}$
Coefficient of linear expansion of steel.		$1.70 \times 10^{-5} \text{ K}^{-1}$
S.T.P		273 K and $1.01 \times 10^5 \text{ Pa}$

Dec 1-1
cent -2
mill -3

Question 1.

- A. The distance from MUST to Catholic University is 10 mi and it takes 20 minutes to drive this distance. Calculate the average speed in meters per second (m/s).

Hint: 1 mi = 1609 m.

[4 marks]

- B. Consider the physical quantities of displacement (s), velocity (v), acceleration (a) and time (t). Determine whether each of the following equations is dimensionally consistent:

i. $s = vt + 0.5at^2$

ii. $s = vt^2$

iii. $v = \sin(at^2/s)$ [6 marks]

- C. A rock is thrown straight up and reaches a height of 10 m. What is the initial velocity of the rock?

[5 marks]

- D. The variation of velocity of a particle with time moving along a straight line is illustrated in **figure 1**. What is the distance travelled by the particle in four seconds?

[5 marks]

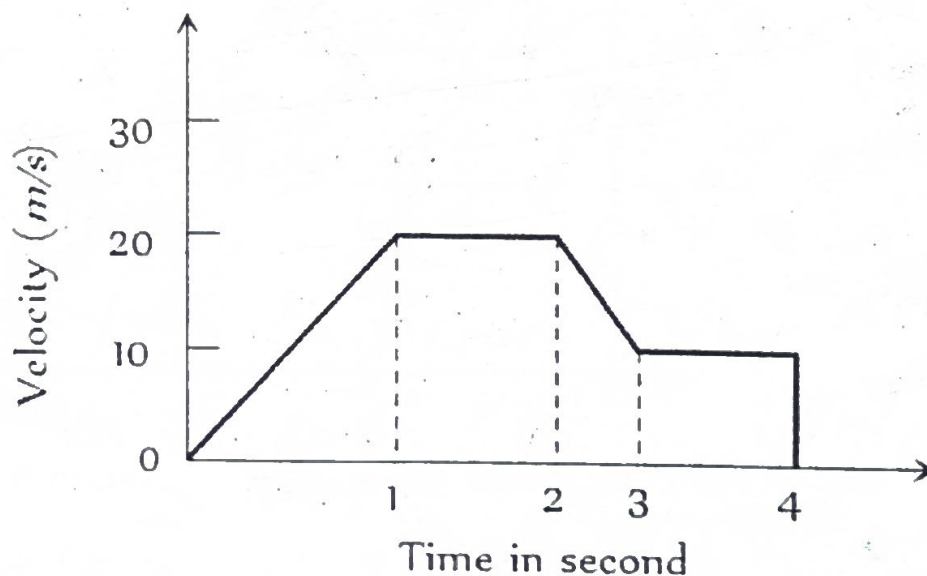


Figure 1

$$s = ut - \frac{1}{2}at^2$$

$$v^2 = u^2 - 2as$$

$$F = ma \quad F = m \frac{v^2}{r}$$

$$p = mv$$

Question 2.

A. State Newton's second law of motion in terms of momentum. [2 marks]

B. A mouse pointer on the display monitor of a computer at its initial position is at point (6.cm, 1.6 cm) with respect to the lower left-side corner. If you move the pointer to an icon located at point (2. cm, 4.5 cm), what is the displacement vector of the pointer? [5 marks]

C. An airplane flying horizontally with a speed of 500 km/h at a height of 800 m drops a crate of supplies as in figure 2. If the parachute fails to open, how far in front of the release point does the crate hit the ground? [5 marks]

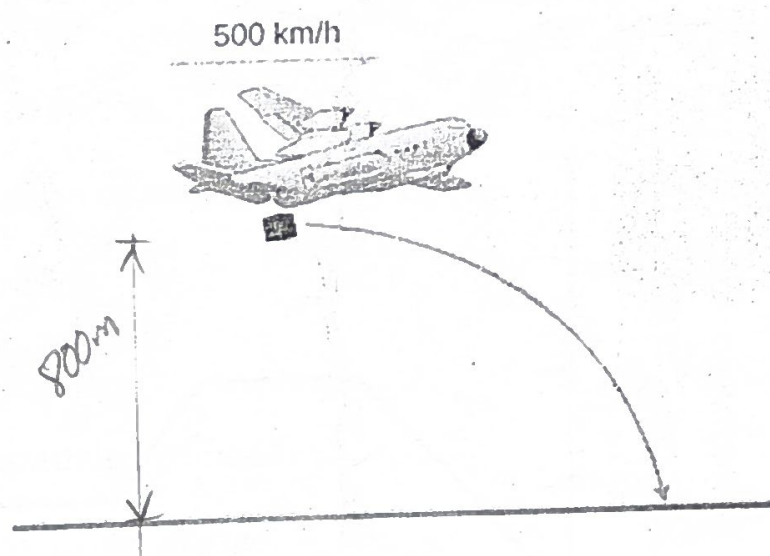


Figure 2.

D. A crate of mass 60 kg is sliding down an incline of an angle 25° . If the coefficient of kinetic friction between the crate and the incline is 0.084, what is its acceleration? [8 marks]

$$v = 138.88888$$

$$19290.12346 = \frac{v^2}{2g} - gt$$

$$-1968.17795$$

$$F = mg \sin \theta$$

$$F = ma$$

$$m a = m g \sin \theta$$

$$s = ut$$

$$v^2 = u^2 - 2gs$$

$$v = u - gt$$

$$h = gt^2$$

Question 3.

A. Define elastic collision

[2 marks]

B. Two objects are connected by a light string passing over a light frictionless pulley as shown in **Figure 3**. The object of mass 5 kg is released from rest. Using the principle of conservation of energy.

i. Determine the speed of the 3-kg object just as the 5-kg object hits the ground.

[3 marks]

ii. Find the maximum height to which the 3-kg object rises.

[3 marks]

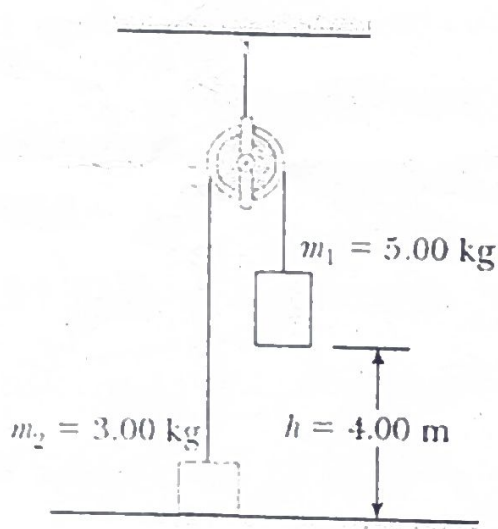


figure 3.

$$v = u + at$$

C. A small car of mass 1200 kg traveling East at 60 km/hr collides at an intersection with a truck of mass 3000 kg that is traveling due North at 40 km/hr. If the collision is perfectly inelastic, what is the velocity of the combined wreckage?

[8 marks]

$$s = ut - \frac{1}{2}gt^2$$

D. Two balls of equal mass undergo an elastic collision in **Figure 4**. Ball 1 is initially traveling horizontally with a speed of 10 m/s, and ball 2 is initially at rest. After the collision, ball 1 moves away with a velocity of 4.7 m/s at an angle of $\theta = 60^\circ$ from its original path and ball 2 moves away at an unknown angle ϕ . Determine the magnitude and direction of velocity of ball 2 after the collision. [9 marks]

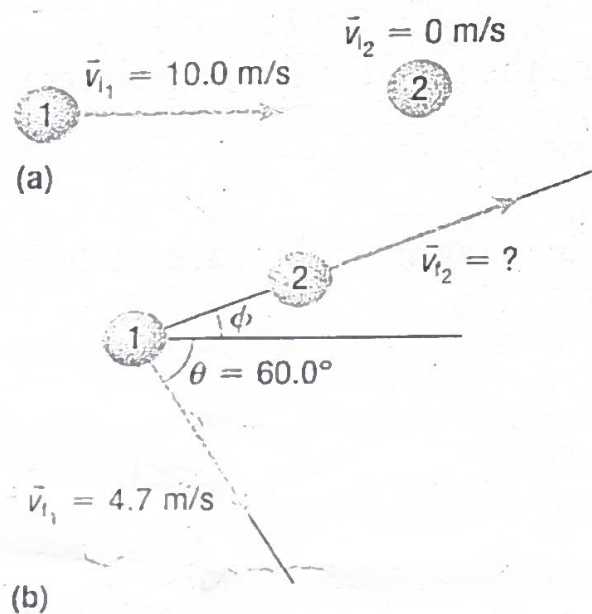
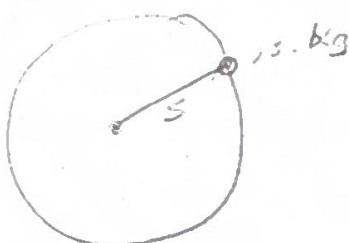


Figure 4

Question 4

- A. A plane flies in a completely horizontal circle at a constant speed of 300 km/h. Calculate the diameter of the circle if the plane's centripetal acceleration is equal to twice the gravitational acceleration. [5 Marks]
- B. A 25.0-kg object is attached to one end of a light, rigid cord. The object rotates in a horizontal circle of radius 5 m at a constant angular speed. The object completes one rotation in 8.5 seconds. Find the tension in the cord.



[5 Marks]

$$\begin{aligned} \theta &= \frac{s}{r} = s = \theta r \\ \omega &= \frac{\theta}{t} = \omega = 10 \\ a &= \frac{v^2}{r} = a = \omega^2 r \end{aligned}$$

Question 5

A. State the following

- i. Zeroth Law of thermodynamics. [2 Marks]
- ii. Absolute zero temperature [2 Marks]
- iii. Triple point of water [2 Marks]
- iv. Latent heat of fusion [2 Marks]
- v. Internal energy [2 Marks]

B. A suspension bridge is constructed using 8 m steel metal bars during winter when the temperature is 2°C . Determine the required spacing between the bars, such that they just come into contact (no compression) during the summer when the temperature is 42°C . The coefficient of thermal linear expansion for steel is $1.2 \times 10^{-5}^{\circ}\text{C}^{-1}$. [5 Marks]

C. Some individuals consider it faster and time-saving to boil water in an electric kettle before transferring it into cookware for subsequent use. If a 2200 W electric kettle is used to heat 1.8 kg of water from 25°C to 90°C , determine the period required to boil the water. Assume water absorbs 100% of the heater's power. [5 Marks]

D. Calculate the energy absorbed from a hand holding a block of ice to first warm 0.30 kg of the block from -5.0°C to 0°C , then to melt the ice, and finally to warm the resulting liquid to 37°C . [5 Marks]

END OF QUESTIONS

5703.84
0.00384
00.0007168